Gawon **Seo**

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Research Interests

World models for embodied agents — when an agent can trust what its world model imagines, how it can improve itself by exploring where that model cannot be trusted, and what a world model should represent for real-time planning.

Education

POSTECH (Pohang University of Science and Technology) *Pohang, S. Korea*
M.S. in Computer Science and Engineering *Feb. 2025 - Feb. 2027 (Expected)*
- Supervised by Prof. Suha Kwak in the Computer Vision Lab

University of Seoul *Seoul, S. Korea*
B.S. in Artificial Intelligence *Mar. 2021 - Feb. 2025*
- Total GPA of 4.09 / 4.5 (**Cum Laude**)

Johannes Gutenberg University Mainz *Mainz, Germany*
Exchange Student Major in Computer Science *Sep. 2023 - Feb. 2024*
- Studied Trustworthy AI

Publications

ACID: Action Consistency via Inverse Dynamics for Planning with World Models **[PDF] [Project Page]**
Gawon Seo, Dongwon Kim, Suha Kwak
Under review at ICLR 2027 · arXiv preprint
- Proposed a decision-time planning framework that enhances the realizability and computational efficiency of action-conditioned world models by introducing Inverse Dynamics-based cycle action consistency.
- Improved planning across 4 action-conditioned world models and 8 tasks including real-robot manipulation.

Planning in 8 Tokens: A Compact Discrete Tokenizer for Latent World Model **[PDF] [Project Page]**
Dongwon Kim, **Gawon Seo**, Jinsung Lee, Minsu Cho, Suha Kwak
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**) 2026*
- Compressed each visual observation into just 8 discrete tokens and built a latent world model on top, achieving $\sim$40$\times$ faster planning with competitive accuracy for real-time robotic control.

Planning in 16 Tokens: A Compact Discrete Tokenizer for Latent World Model
Dongwon Kim, Jinsung Lee, **Gawon Seo**, Minsu Cho, Suha Kwak
*The Conference on Robot Learning Workshop (**CoRLW**) - Learning to Simulate Robot Worlds, 2025*

Enhancing Visual Classification using Comparative Descriptors **[PDF]**
Hankyeol Lee, **Gawon Seo**, Wonseok Choi, Geunyoung Jung, Kyungwoo Song, Jiyoung Jung
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**) 2025*
- Proposed Comparative Descriptors for VLMs to enhance image classification by emphasizing subtle differences between visually similar classes.

Generating Accurate and Detailed Captions for High-Resolution Images **[PDF]**
Hankyeol Lee, **Gawon Seo**, Kyounggyu Lee, Dogun Kim, Kyungwoo Song, Jiyoung Jung
arXiv preprint
- Enhanced VLM captioning accuracy via LLM-based object verification and region-specific refinement.

Experience

Computer Vision Lab, POSTECH ***[Lab homepage]***
Research and Teaching Assistant *Feb. 2025 - Present*
- Advisor: Prof. Suha Kwak
- Working on efficient and realizable planning with World Models for embodied AI.
- Collected teleoperated manipulation demonstrations and deployed learned policies on an AgileX PiPER 6-DoF arm.
- Publications: ACID (under review at ICLR 2027), CompACT (CVPR 2026)

Robotics and Computer Vision Lab, University of Seoul ***[Lab homepage]***
Undergraduate Researcher *Apr. 2024 - Jan. 2025*
- Advisor: Prof. Jiyoung Jung
- Worked on grounding Vision-Language Models in what they actually see.
- Publications: Comparative Descriptors (WACV 2025), High-Resolution Image Captioning (arXiv preprint)

One Eleven *Seoul, S. Korea*
AI Developer *Jun. 2023 - Jul. 2023*

- Developed a recommendation system to prioritize store locations to install power bank rental machines, analyzing commercial and GIS data.

Geo DataBase Lab *University of Seoul*
Undergraduate Researcher *Jul. 2022 - Feb. 2023*

- Advisor: Prof. Chulmin Jeon
- Participated in the research "A Model for Evaluating Good Two-Wheeler Drivers Using AHP Analysis Technique."

Skills

Machine Learning Python, PyTorch, Behavior Cloning, Diffusion Policy, World Models, MPC/CEM Planning
Robotics ROS, LeRobot, MuJoCo, NVIDIA FleX, Franka, AgileX PiPER, Teleoperation Setup, Real-World Robot Evaluation Design

Awards & Honors

2025.02 **CUM LAUDE,** Awarded the CUM LAUDE for excellent academic records
2024.11 **AICOSS President's Award,** Developed Real-time gaze tracker for Barrier-free kiosk, 2024 CO-SHOW AIM Challenge
2024.11 **Creativity Award,** Developed data-driven public value solutions, 2024 K-DataScience Hackathon
2024.09 **2nd Place in 2D Semantic Segmentation,** 2024 Autonomous Driving AI Algorithm Development Challenge
2023.01-2024.12 **Dongyang E&P Honors,** Awarded to Engineering students with high achievements
2023.02 **AICOSS President's Award,** AICOSS Industry-University Linkage Deep Learning Competition
2022.09 **Department Honors,** Awarded to undergraduate researchers with high achievements
2022.03 **Academic Awards,** Awarded to students with high achievements throughout the semester
2021.09 - 2022.06 **Semester High Honors,** Awarded to students with high achievements throughout the semester

Teaching Experience

- CSED537/AIGS537, Artificial Intelligence & Data Science (Spring 2026)

Extracurricular Activity

Inter-university IT association TAVE *Seoul, S. Korea*
Deep Learning Team Member and Presenter *Mar. 2023 - Jul. 2023*

- Learned Computer Vision theory of classical and deep learning methods and implementation.
- Participated in the court decision prediction AI team competition and presented the project.

English Speech Club *Seoul, S. Korea*
Member *Mar. 2023 - Jul. 2023*

- Participated in regular gatherings with international students at the University of Seoul to discuss diverse topics.

App Development Team UOSLIFE *Seoul, S. Korea*
Backend Developer *Mar. 2022 - Dec. 2022*

- Built foundational knowledge in backend, database, and algorithms through technical studies to support the operation of a University of Seoul community app 'UOSLIFE' providing essential student services.
- Developed comprehensive communication skills by participating in both internal developer meetings and cross-functional alignment sessions with product planning and design teams.

References

Prof. Suha Kwak, Associate Professor *POSTECH*
Academic Advisor

- Email: suha.kwak@postech.ac.kr
- Homepage: https://suhakwak.github.io

Prof. Jiyoung Jung, Associate Professor *University of Seoul*
Academic Advisor

- Email: jyjung@uos.ac.kr
- Homepage: https://sites.google.com/view/rcvuos/home